AIVOA.AI

Software Testing & QA Challenge

Submission pack
QMS understanding · Deviation workflow · Bug report · Product improvements · Video plan

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Module: Quality Events → Deviation → New Record

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1. Executive Summary

This report explains how a Quality Management System (QMS) supports an Active Pharmaceutical Ingredient (API) manufacturing process, with particular focus on the path from **deviation reporting through investigation, root cause, CAPA, and closure**.

It also outlines a user-centred test approach for AIVOA.AI's **New Deviation Record** workflow. The aim is to assess whether a quality user can create and manage a clear, complete, traceable record—not simply whether individual fields and buttons work.

Testing status: The bug examples included in the earlier draft are not verified evidence. They are listed below as scenarios to test. Confirm each in the application and record the actual result before submitting it as a bug.

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2. QMS Understanding: API Manufacturing Example

What is a QMS?

A **Quality Management System** is the set of processes, responsibilities, records, and controls an organization uses to consistently make products that meet defined quality requirements.

In API manufacturing, a QMS helps teams:

- Protect patients by identifying and controlling quality risks.
- Follow approved procedures and applicable GMP requirements.
- Record what happened, who took action, and when.
- Investigate problems and prevent them from recurring.
- Make evidence-based decisions about materials, processes, and batch disposition.

A QMS is not just software or a collection of forms. It is the operating framework that connects people, procedures, equipment, materials, and records.

Example: API batch lifecycle

Imagine a facility manufacturing an API. During a reactor step, the process temperature briefly exceeds its approved operating range.

Stage	What happens	Who commonly uses it	Why it matters
□ Raw materials	Materials arrive, are identified, sampled, tested, and released or rejected. Supplier and lot information is recorded.	Receiving, QC, QA, Supplier Quality	Prevents unsuitable or unverified material from entering production.
⚙ Manufacturing	Trained operators follow the approved manufacturing instructions and record the actual steps and process conditions.	Production, supervisors, QA	Provides evidence that the process was performed as approved.
🔍 In-process checks	Operators or QC check parameters such as temperature, pH, time, or appearance against approved limits.	Production, QC, QA	Helps detect process problems before they affec the final product.
⚠ Deviation	An unexpected event or departure from an approved instruction or limit is reported and documented.	Operator, supervisor, QA	Preserves the facts and prompts appropriate containment and assessment.

Stage	What happens	Who commonly uses it	Why it matters
🕵️ Investigation	The team gathers records, interviews relevant staff, examines equipment and materials, and assesses impact.	Investigator, Production, QC, Engineering, QA	Determines what happened and what product o process may be affected.
🎯 Root cause	The investigation identifies the underlying cause, supported by evidence—not merely a symptom or assumption.	Investigator, subject-matter experts, QA	Helps ensure actions address the actual cause.
🛠️ CAPA	Corrective action addresses the current issue; preventive action reduces the chance of recurrence.	Action owners, QA, relevant departments	Converts investigation conclusions into managed and verifiable action.
✅ Closure and batch disposition	QA checks that required documentation, impact assessment, actions, and approvals are complete. Batch release is a separate quality decision based on all relevant evidence.	QA, authorized quality approver	Ensures the record is complete and supports a justified batch decision.

Stage	What happens	Who commonly uses it	Why it matters
📣 Complaint or recall	A complaint is assessed and connected to relevant product and batch records. A recall may be considered if risk and evidence warrant it.	Customer Quality, QA, Regulatory, leadership	Connects information from the market back to manufacturing and patient risk.
🤝 Supplier Quality	If a material or supplier may have contributed, the issue can inform supplier investigation, monitoring, corrective action, or qualification decisions.	Supplier Quality, Procurement, QA	Prevents supplier-related problems from repeating across batches.

Deviation example

During a reactor step, the approved temperature range is exceeded for five minutes. The operator records the event and promptly informs the supervisor and QA. The affected material and batch are assessed according to site procedures while the team preserves relevant records, including equipment readings and batch documentation.

The deviation record should capture the event factually: **what happened, when and where it happened, what was affected, and what immediate actions were taken**. It should not assume the cause before investigation.

Investigation → Root Cause → CAPA → Closure

1. **Report and contain:** Record the event promptly. Take appropriate immediate steps to protect people, product, and data, following approved procedures.
2. **Assess impact and risk:** Determine what materials, batches, process steps, equipment, records, or other products may be affected.

3. **Investigate:** Gather objective evidence, such as batch records, equipment history, calibration status, alarms, interviews, and relevant procedures.
4. **Identify root cause:** Use a structured method, such as 5 Whys or a cause-and-effect analysis, where appropriate. Distinguish confirmed evidence from assumptions.
5. **Define CAPA:** Assign actions, owners, due dates, and completion evidence. Actions should address the supported cause and the affected scope.
6. **Verify effectiveness:** After implementation, assess whether the actions prevented recurrence using defined, appropriate evidence.
7. **Close with review:** Confirm the investigation, impact assessment, actions, approvals, and supporting records are complete. Closure should be authorized according to the organization's procedures.

Key distinction: Correcting the immediate issue is not necessarily the same as eliminating its root cause. Likewise, an action being completed does not automatically prove it was effective.

How related QMS processes connect

- **Complaints** may reveal an issue that was not detected during manufacturing. The complaint should be assessed and linked, where appropriate, to product, batch, deviation, and investigation records.
- **Recalls** are risk-based decisions made through the organization's authorized procedures. A deviation or complaint may contribute evidence, but does not automatically mean a recall is required.
- **Adverse events** are safety reports associated with a product and are handled through applicable pharmacovigilance and regulatory processes. They may require coordination with quality teams, but are not interchangeable with a deviation.
- **Supplier management** connects incoming-material issues to supplier performance, investigations, supplier corrective actions, audits, and qualification decisions.

What the AIVOA.AI deviation workflow is intended to achieve

A deviation workflow should help users create a consistent, reviewable record of an unexpected event and route it through the right assessment and approval steps. A useful workflow supports clear ownership, timely investigation, evidence-based conclusions, CAPA tracking, and an auditable closure decision.

The application should make the record's status and next required action understandable to the user while helping preserve traceability throughout the event lifecycle.

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3. Testing the New Deviation Record

User expectation

As a Quality user in a pharmaceutical company, I would expect the workflow to:

- Make clear which information is required and why.
- Prevent incomplete or invalid submissions where required.
- Preserve entered information if I save, navigate away, or reopen the record.
- Clearly distinguish draft, submitted, under-review, and closed states, if those states are supported.
- Show meaningful confirmation or actionable validation messages.
- Avoid creating duplicate records when I click repeatedly or retry after a delay.
- Preserve a traceable history of important actions and changes.

Suggested test coverage

Area	Example checks
Required fields	Submit with required information missing; check that errors identify the fields and how to fix them.
Field validation	Test invalid formats, whitespace-only values, excessively long text, and unusual characters where relevant.
Save and reopen	Save a draft or record, leave the page, and reopen it. Confirm values and status are preserved.
Refresh and navigation	Refresh or navigate away with unsaved changes. Check whether the user is warned and whether saved data remains intact.
Repeated actions	Click save or submit repeatedly; retry after a slow response. Check for duplicate records or confusing status.
Workflow sequence	Check that actions are available only when their prerequisites are met and that the user can understand the next step.

Area	Example checks
Usability	Check labels, field instructions, error placement, keyboard navigation, and readability.
Traceability	Where available, check record identifiers, timestamps, ownership, status history, and audit information.

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4. Bug Report

Verification status

The following are **test scenarios, not confirmed defects**. Reproduce them in the application and replace the “Actual result” text with what you observe. Capture screenshots or a short recording and note the date, browser, and steps.

Candidate issue 1: Saved data may not persist after refresh

- **Steps:** Open **Quality Events → Deviation → New Record**. Enter valid data and save. Refresh the page or navigate away and reopen the record.
- **Expected:** The saved record and its values remain available, with a clear status or confirmation.
- **Actual: Not verified. Record observed result here.**
- **Why this matters:** Losing a user’s saved information could cause rework, delay reporting, or leave a quality event undocumented.

Candidate issue 2: Required-field validation may be insufficient

- **Steps:** Open a new deviation and attempt to save or submit with required fields blank or containing only spaces.
- **Expected:** The application should prevent an invalid submission and clearly identify each missing or invalid field.
- **Actual: Not verified. Record observed result here.**
- **Why this matters:** A record without essential information may be difficult to triage or investigate and may create avoidable follow-up work.

Candidate issue 3: Navigation may risk losing work or context

- **Steps:** Enter information in the record. Use the application's Back or navigation controls before saving, then return to the module.
- **Expected:** The application should preserve saved information and warn the user before discarding unsaved changes.
- **Actual: Not verified. Record observed result here.**
- **Why this matters:** Users should not unknowingly lose work or be left unsure whether a record was saved.

Evidence to capture for each confirmed bug

- Bug title and unique reference.
- Environment: browser, operating system, and test date.
- Exact reproduction steps and test data that does not contain sensitive information.
- Expected and actual results.
- Screenshot or recording, if permitted.
- Frequency: consistent, intermittent, or not yet determined.
- User and quality impact, without overstating regulatory conclusions.

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5. Product Improvements: 10X Opportunities

1. **□ AI-supported investigation assistance**
   Suggest relevant investigation questions or similar historical cases based on the event details. Clearly show the source and uncertainty, and require a human reviewer to accept or reject suggestions.
2. 📎 **Contextual traceability**
   Link deviations to relevant batches, materials, suppliers, equipment, procedures, complaints, and CAPAs. Make these relationships easy to review without duplicating records.
3. 📈 **Recurrence and trend detection**
   Highlight recurring event patterns by equipment, process, material, or cause category. Let authorized users review the evidence before creating actions or escalating an issue.
4. 👥 **Collaborative investigation workspace**
   Support clear action ownership, comments, evidence attachments, and notifications

so cross-functional teams can collaborate in one record rather than relying on scattered email threads.

5. ⏰ **Risk-aware task and deadline management**
   Provide configurable due dates, reminders, escalation rules, and a dashboard for overdue or at-risk records. Keep the status and next action visible to each responsible user.

6. □ **Clear audit trail and change history**
   Make it easy to see who changed key information, when it changed, and—where appropriate—why. Present the history in a readable timeline.

7. ✅ **CAPA effectiveness tracking**
   Help define effectiveness checks, evidence, owners, and review dates when a CAPA is created, then prompt for the check before the record is closed where the workflow requires it.

8. □ **Guided, role-aware workflows**
   Use contextual instructions and progressive steps to help users understand what they need to enter and what happens next, while tailoring tasks to their role and permissions.

9. 📊 **Quality insights with drill-down**
   Provide dashboards for event types, aging, recurrence, and closure trends. Each metric should link back to the records behind it so users can verify the data.

10. 🌐 **Complaint and supplier feedback loops**
    Help authorized users connect complaint and supplier-quality information to relevant events and batches, supporting a more complete view of potential impact and recurrence.